

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI

Semester: 5

Academic Year: 2026-27

Division: E

Batch: 1

Practical No: 6

Roll no: 52

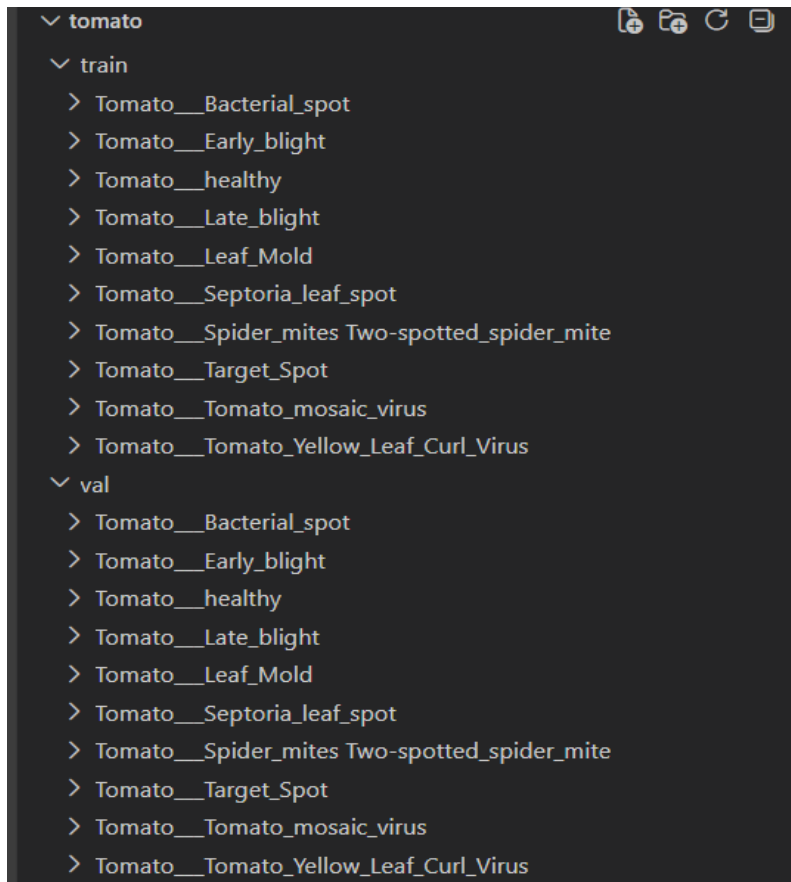
PRN No: 12410371

Name of Student: Vaidik Samarth

Subject Name: Deep Learning

Problem Statement: Design and implement a Convolutional Neural Network (CNN) for image classification using the Tomato or Soybean disease dataset.

Code and Output:



```

from tensorflow.keras import layers, models

model = models.Sequential([

    # Input + normalization
    layers.Rescaling(
        1./255,
        input_shape=(128, 128, 3)
    ),

    # CNN Block 1
    layers.Conv2D(32, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),

    # CNN Block 2
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),

    # CNN Block 3
    layers.Conv2D(128, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),

```

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 128, 128, 3)	0
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

Total params: 3,305,930 (12.61 MB)

Trainable params: 3,305,930 (12.61 MB)

Non-trainable params: 0 (0.00 B)

```
model.compile(  
    optimizer='adam',  
    loss='sparse_categorical_crossentropy',  
    metrics=['accuracy']  
)
```

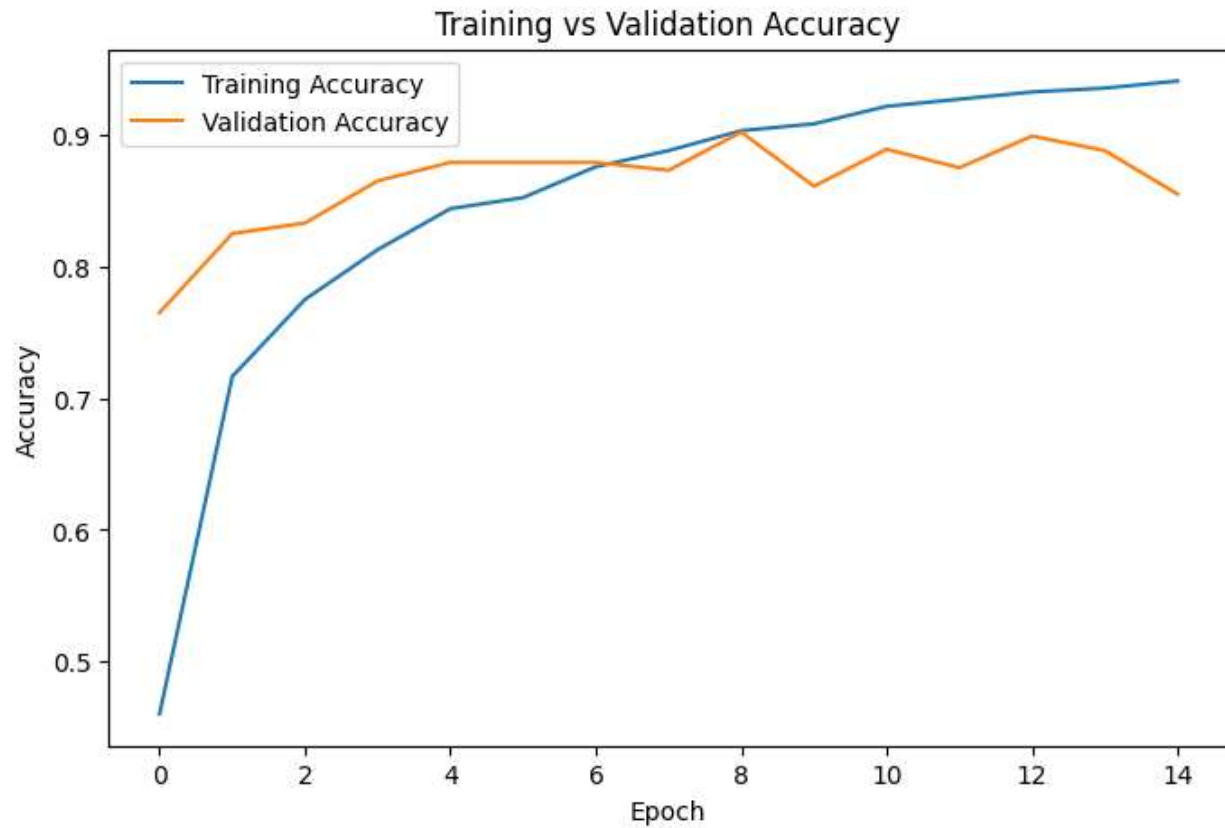
```
history = model.fit(  
    train_dataset,  
    validation_data=val_dataset,  
    epochs=15  
)
```

```
val_loss, val_accuracy = model.evaluate(val_dataset)
```

```
print("Validation Loss:", val_loss)  
print("Validation Accuracy:", val_accuracy)
```

```
32/32 ————— 1s 17ms/step - accuracy: 0.8550 - loss: 0.5064  
Validation Loss: 0.5063943862915039  
Validation Accuracy: 0.8550000190734863
```

```
plt.figure(figsize=(8, 5))  
  
plt.plot(  
    history.history['accuracy'],  
    label='Training Accuracy'  
)  
  
plt.plot(  
    history.history['val_accuracy'],  
    label='Validation Accuracy'  
)  
  
plt.xlabel("Epoch")  
plt.ylabel("Accuracy")  
plt.title("Training vs Validation Accuracy")  
plt.legend()  
  
plt.show()
```



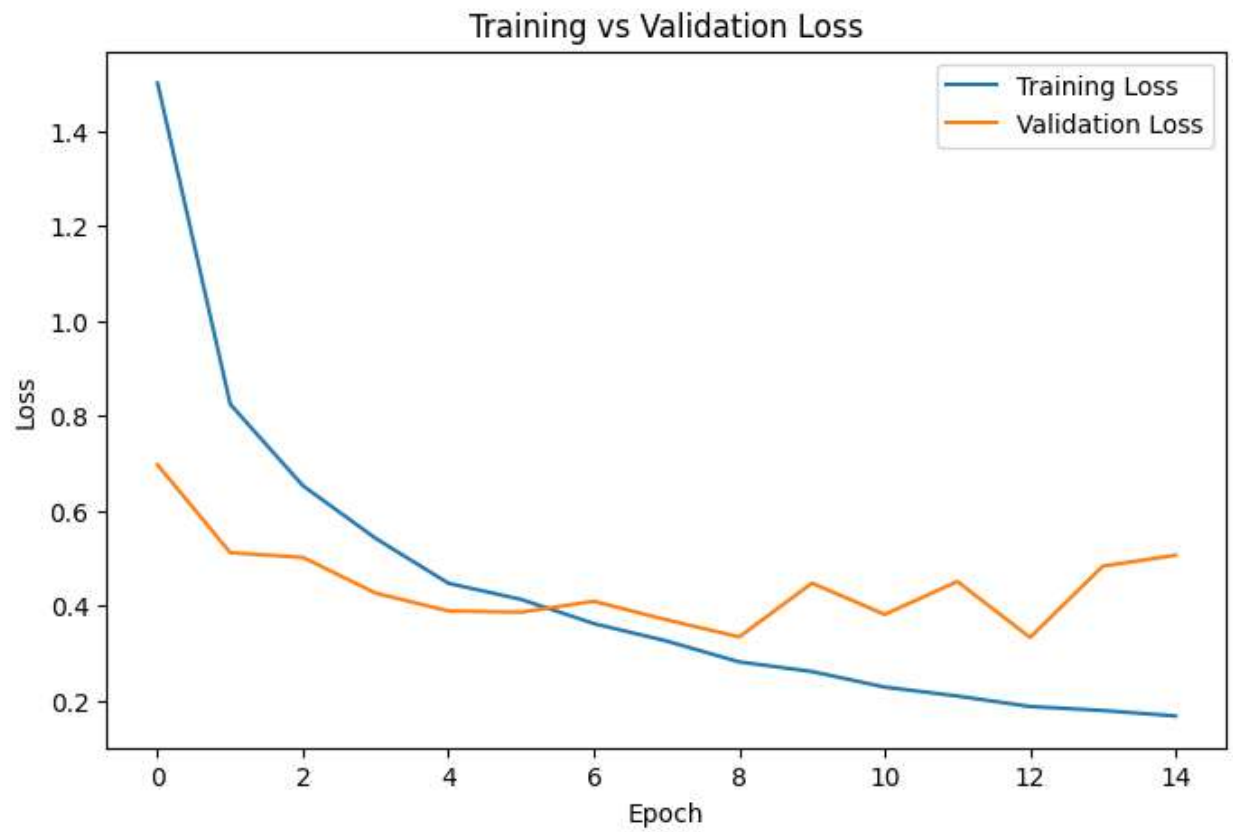
```
plt.figure(figsize=(8, 5))

plt.plot(
    history.history['loss'],
    label='Training Loss'
)

plt.plot(
    history.history['val_loss'],
    label='Validation Loss'
)

plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()

plt.show()
```



```
model.save("tomato_disease_cnn.keras")
```